An illustrated guide

Rain from Orbit

An Illustrated Guide to GPM's Dual-Frequency Precipitation Radar



Built on the GPM/DPR Level-2 Algorithm Theoretical Basis Document.
Every number in this book is computed from Mie scattering theory, not fitted.

Why a radar in orbit, and the constellation it calibrates

02 **The Ambiguity at the Heart of It**
Three drops in a box, and the measurement that cannot tell them apart

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06 **What Kind of Rain Is This?**
The bright band, and three ways to read a profile

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08 **The Monsoon Problem**
Where every assumption in this book is tested at once

Pages marked Now You Try It are interactive simulators in the online edition. Here they are printed as a snapshot of one setting.

CHAPTER 01

A Planet's Worth of Rain!!!

How do you weigh all the rain falling on Earth, when almost none of it falls on a rain gauge?

How This Guide Works

1 Every page has **one header bar** like the green one above, telling you exactly which single idea the page is about. Nothing else happens on that page.

Inside, follow the **circled numbers in order**. They run from low to high across the whole chapter, and each one depends on the one before it.

2 Two characters will argue their way through all of this. Meet them now, because the whole problem is hiding in the gap between what they each know.



Dot — a raindrop. I know *exactly* what I am: 2 millimetres across, falling at 6 metres a second, and I know precisely how many sisters I have in this cubic metre. Ask me anything.

Core — the GPM Core Observatory, 442 km up. I know none of that. I know exactly one number: how much of the pulse I sent down came back. Everything else, I have to work out.



3 Three more things you will see over and over:

SPLAT!!! appears when an idea finally lands. **TERMINOLOGY ALERT!!!** boxes appear when a word is about to be used in two different ways by two different papers. And purple boxes are **simulators** — real physics, computed live, that you can grab and move.

Psst! The colours mean things, all the way through. Amber is always Ku-band. Teal is always Ka-band. Blue is whatever we are trying to solve for, and red is what the radar actually measured.

Measuring Rain From Space: Main Ideas

4



Hang on, **Core**. Why is this hard? Put a bucket outside. Measure what's in the bucket. Rain measured!

That works for the bucket, **Dot**. It tells you nothing about the next valley, and nothing at all about the ocean — which is seventy percent of the planet and has essentially no buckets.



5 **The problem:** rain is both violently local and globally consequential. One thunderstorm cell three kilometres across can drop more water in an hour than the surrounding hundred kilometres gets in a week.

So this is not a measurement problem. It is a **sampling** problem — and the instruments are not where the water is.

6 **A solution:** look down at it. But *how* you look down matters enormously, and there are two very different ways.

- A **radiometer** listens to the microwaves the atmosphere emits and scatters. Wide swath, whole globe, several times a day.
- A **radar** sends its own pulse and times the echo.

7 The difference is everything, and it is the subject of the next page.

SPLAT!!!

Then we will meet the satellite that does both at once.

Why a Radar: Details Part 1

8 A **radiometer** measures a brightness temperature — one number, summed through the entire column of atmosphere beneath it.

I can see *that* there is water down there. I cannot see whether it is at 2 km or at 10 km, whether it is frozen or melted, or where it changes from one to the other.



9 A **radar** transmits a pulse and times the echo. And timing is the whole trick:

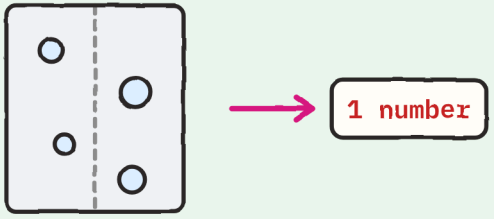
`every echo arrives with its own` **`range`** `attached`

...and range, from a satellite looking down, is height.

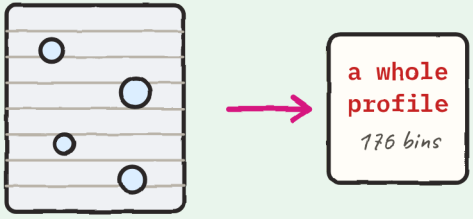
So a radar does not return a picture of a storm. It returns a **vertical slice through it**, kilometre after kilometre along the orbit track.

10

RADIOMETER — one number per column



RADAR — one number per range bin



Same storm, same moment. The radiometer compresses it; the radar resolves it.

The Core Observatory: Details Part 2

(11) The first spaceborne precipitation radar flew on **TRMM** in November 1997 and covered the tropics. Its successor, the **GPM Core Observatory**, launched from Tanegashima on **27 February 2014** — a joint NASA and JAXA mission.

It carries two instruments at once: **GMI**, a radiometer with a 904 km swath, and **DPR**, the Dual-frequency Precipitation Radar.

(12) Now the orbit, which is a deliberate and slightly unusual choice:

- **65° inclination** — reaches well past the tropics into the mid-latitudes, where snow and shallow frontal rain live.
- **Non-sun-synchronous** — the crossing time drifts.
- **~93 minutes** per orbit, about sixteen orbits a day.

(13)



Why would you *want* the crossing time to drift? That sounds like sloppy engineering.

Because a sun-synchronous orbit would see the tropics at 01:30 and 13:30 and *never any other time*. Convection over land peaks in the afternoon. Offshore convection peaks at night. I would have been permanently blind to the daily cycle of the thing I was built to measure.



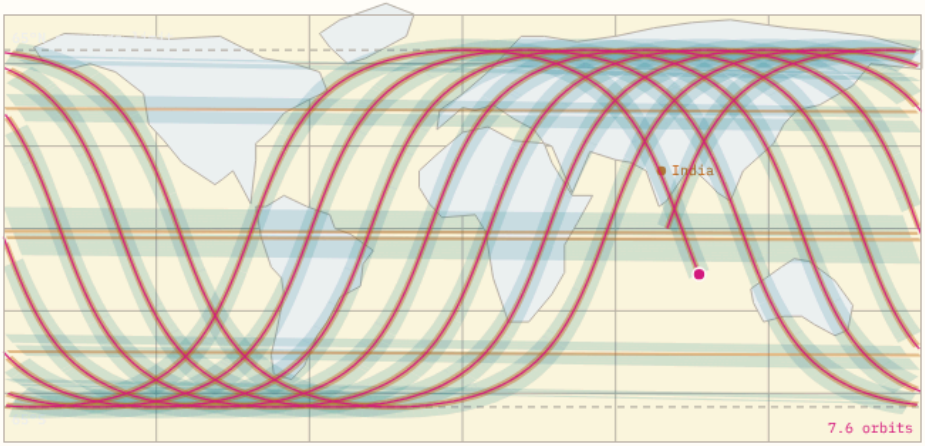
Over a month, the drift lets GPM sample every hour of the day at every latitude it reaches. SPLAT.

NOTE: In November 2023 the spacecraft was boosted from about 407 km to 442 km to slow fuel use. The footprint grew from 5.0 km to 5.5 km and the swath from 256 km to 278 km. It is expected to keep observing into about 2030.

Now You Try It: One Day of Orbits

One day of orbits

drag the slider, or press play



ground track DPR swath (≈245 km) GMI swath (904 km) dashed: 65° coverage limit

Elapsed time 11 h 40 m

▶ Play 24 h Show swaths

14 Run the full day. Then look at the **white space between the amber swaths**.

That is my problem in one picture. I pass over any given place roughly twice a day, and a thunderstorm lives for about an hour. I will miss almost everything.

The Constellation: Details Part 3

(15) So one radar is not the global product. It is the **ruler that calibrates one**.

Because GMI and DPR fly on the same platform, every overpass gives a matched pair: what a radiometer sees, and what the radar underneath it knows is actually there.

(16) Those pairs train the retrieval used on every other radiometer in the **GPM constellation** — DMSP's SSMIS, GCOM-W1's AMSR-2, MetOp's MHS, and the ATMS instruments on NOAA-20, NOAA-21 and Suomi-NPP.

None of those carries a radar. Every one of them borrows Core's.

(17) The merged product is **IMERG** — Integrated Multi-satellitE Retrievals for GPM.

PROPERTY	VALUE
Grid	`0.1° — about 11 km at the equator`
Cadence	`half-hourly`
Coverage	`60°N–60°S full; 90°N–90°S microwave-only`
Early run	`≈ 4 h latency — operational use`
Late run	`≈ 12–14 h`
Final run	`≈ 3.5 months, after gauge analysis`

WHY THIS MATTERS!!!

Everything downstream rests on the radar being right. If Core's rain rates are biased, that bias propagates into the radiometer training, then into IMERG, then into every flood model, crop forecast and reanalysis that ingests it.

The rest of this guide is about why getting the radar right is so much harder than pointing it down.

Chapter 1: Self-Check

GPM flies a non-sun-synchronous orbit. What does that buy the mission?

- **A** It samples every local time of day at a given latitude over the course of a month, so the diurnal cycle of precipitation is resolved.
- **B** It keeps the solar panels permanently illuminated, which simplifies the power budget.
- **C** It holds the satellite over the tropics, where most precipitation falls.
- **D** It reduces the fuel needed to maintain altitude against atmospheric drag.

Answer: A. A sun-synchronous orbit crosses a given latitude at the same two local times forever, so it would be systematically blind to the daily cycle of convection — afternoon storms over land, nocturnal maxima offshore. The drifting crossing time is what lets GPM build an unbiased diurnal climatology.

What is the Core Observatory's role for the rest of the constellation?

- **A** It is the calibration reference: its radar–radiometer pairs train the retrievals applied to partner radiometers that carry no radar.
- **B** It relays the partner satellites' data to the ground through TDRS.
- **C** It is the only satellite in the constellation that measures precipitation; the others measure water vapour.
- **D** It fills the coverage gaps left by the partner satellites' narrower swaths.

Answer: A. The partner satellites — SSMIS, AMSR-2, MHS, ATMS — are passive radiometers flying their own missions. None has a radar. GPM's value is that it sees both the radiometer signal and the radar truth at the same time and place, which is what makes a uniform multi-satellite product like IMERG possible.

CHAPTER 02

The Ambiguity at the Heart of It!!!

A radar does not measure rainfall. It measures something related to rainfall — and "related" is doing an enormous amount of work in that sentence.

The Ambiguity: Main Ideas

1

Dot, I sent a pulse down and 29.0 dBZ came back. Tell me how hard it is raining.





Can't. Not from that number alone. It could be half a millimetre an hour. It could be more than five.

...those differ by a factor of eleven.



2

The problem: the echo and the rain rate are two *different* summaries of the same population of drops, and they weight that population very differently.

- The **echo** cares about D^6.
- The **rain rate** cares about roughly $D^{3.7}$.

Nearly double the power. That gap is the whole chapter.

3

How we are going to prove it: not with an integral. With **three raindrops in a box**, and arithmetic you can check on paper.

Then we will build a second box that gives Core *exactly the same echo* and rains eleven times harder.

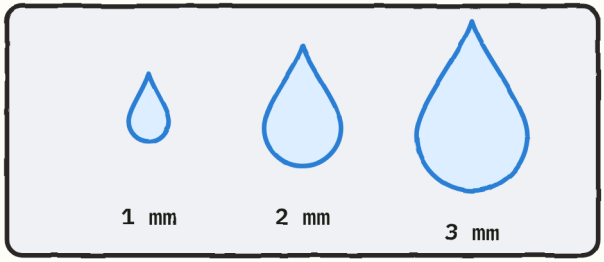
Psst! Real rain has thousands of drops per cubic metre. We are using three so the numbers stay small enough to follow. Josh would approve.

Three Drops in a Box: Details Part 1

4 Here is **Box A**. It is one cubic metre of air containing exactly three raindrops, of diameter **1 mm**, **2 mm** and **3 mm**.

That's me in the middle. Two millimetres. Pleased to meet you.

BOX A — one cubic metre



Drawn to scale. Notice how much bigger the 3 mm drop is — hold on to that.

5 Two formulas, and that is all the physics we need for now. **Reflectivity** is the sum of the sixth powers of the diameters:

$$Z = \Sigma\, D^6 \quad [\mathrm{mm^6\ m^{-3}}]$$ — *sum over every drop in the box*

And **rain rate** is the volume of water passing through, so drop volume times how fast it falls:

$$R = 6\pi \times 10^{-4} \times \Sigma\, D^3\, v(D) \quad [\mathrm{mm\ h^{-1}}]$$

where $v(D) = 3.78 \times D^{0.67}$ *metres per second — bigger drops fall faster*

NOTE: Why D^6? Because for drops much smaller than the wavelength, scattering is **Rayleigh**, and the backscatter cross-section of a sphere goes as the sixth power of its diameter. Chapter 5 is about what happens when that stops being true.

Computing Z by Hand: Details Part 2

6 Raise each diameter to the sixth power and add them up. That is literally all.

$$Z = 1^6 + 2^6 + 3^6$$
$$Z = 1 + 64 + 729$$
$$Z = \mathbf{794}\ \text{mm}^6\ \text{m}^{-3}$$

7 That number spans about fifteen orders of magnitude between drizzle and hail, so nobody ever quotes it raw. Convert to decibels:

$$\text{dBZ} = 10 \log_{10}(794)$$
$$= \mathbf{29.0\ \text{dBZ}}$$

This is the number Core measured. It is the *only* number Core measured.

8 Now look at where that 794 came from:

DROP	D^6	SHARE OF Z
1 mm	1	0.1 %
2 mm	64	8.1 %
3 mm	729	91.8 %

One drop out of three is making 92% of the echo.

9

So when I measure 29.0 dBZ, I am almost entirely measuring your biggest sister. The 1 mm drop may as well not be in the box.





Right. But she still *rains*. Watch what happens when we compute R.

Computing R by Hand: Details Part 3

10 Same three drops. Now cube each diameter, multiply by that drop's fall speed, and add.

$v(1) = 3.78 \times 1^{0.67} = 3.78$ m/s → $D^3v = 1 \times 3.78 = 3.78$

$v(2) = 3.78 \times 2^{0.67} = 6.01$ m/s → $D^3v = 8 \times 6.01 = 48.11$

$v(3) = 3.78 \times 3^{0.67} = 7.89$ m/s → $D^3v = 27 \times 7.89 = 213.07$

$\Sigma = 265.0$

$R = 6\pi \times 10^{-4} \times 265.0 = 0.50$ mm h^{-1}

Half a millimetre an hour. Light drizzle.

11 Now compare the two sums, drop by drop. Take the **3 mm** drop against the **1 mm** drop:

echo ratio: 729 / 1 = **729 ×**

rain ratio: 213.07 / 3.78 = **56 ×**

12 Read that again. Make a drop **three times wider** and it returns **729 times** the echo — but carries only **56 times** the rain.

SPLAT!!!

Big drops are loud and small drops are quiet, but small drops still get you wet. Core only hears the loud ones.

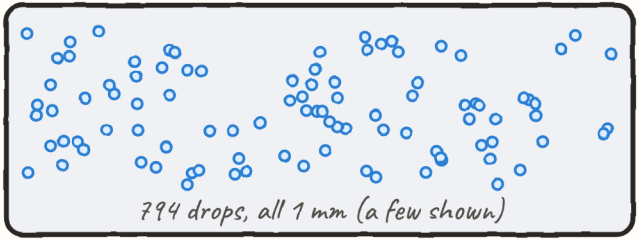
The Second Box: Details Part 4

13 Let us now build **Box B** on purpose, to fool Core.

The rule is simple: get Z to come out to exactly **794** again, using only 1 mm drops. Since each 1 mm drop contributes $1^6 = 1$, we need...

$$Z = N \times 1^6 = 794 \quad \rightarrow \quad N = \mathbf{794\ drops}$$

BOX B — the same cubic metre



14 Core's measurement is **identical**. Same 794, same 29.0 dBZ. Now compute the rain rate the same way we did before:

$$\Sigma\, D^3 v = 794 \times (1 \times 3.78) = \mathbf{3001.3}$$

$$R = 6\pi \times 10^{-4} \times 3001.3 = \mathbf{5.66\ mm\ h^{-1}}$$

15

	BOX A	BOX B
What Core measures	29.0 dBZ	29.0 dBZ
What is actually falling	0.50 mm/h	5.66 mm/h
Liquid water	0.019 g/m³	0.416 g/m³

DOUBLE SPLAT!!!

Eleven times the rain. Twenty-two times the water. And I cannot tell these two boxes apart *at all*.



Two Numbers, Not One: Details Part 5

16 So what actually differs between the boxes? Two things, and the whole of GPM's Level-2 algorithm is built to find them.

How big the drops are, and **how many there are**. Box A is *few and large*. Box B is *many and small*.

TERMINOLOGY ALERT!!!

D_m, the *mass-weighted mean diameter*, is not the average diameter. It is $\Sigma D^4 / \Sigma D^3$ — the size of the drop carrying the average bit of water. It is weighted toward the big ones, which is what you want, because the big ones carry the water.

N_w is a *normalised* concentration. Loosely: how many drops there would be if they were all D_m-sized. It is defined so that it does not drag the size information along with it.

18 Compute D_m for Box A — we already have the pieces:

$\Sigma D^4 = 1 + 16 + 81 = 98$ $\Sigma D^3 = 1 + 8 + 27 = 36$

$D_m = 98 / 36 =$ **2.72 mm** — *even though two of the three drops are smaller than that*

And for Box B every drop is 1 mm, so D_m = **1.00 mm** exactly.

	D_M	$\text{LOG}_{10}\ N_W$	Z	R
Box A	2.72 mm	1.45	29.0 dBZ	0.50 mm/h
Box B	1.00 mm	4.53	29.0 dBZ	5.66 mm/h

One measurement. Two unknowns. That is an underdetermined system, and no amount of calibration fixes it.

From Three Drops to Real Rain: Details Part 6

19 Real rain does not come in three neat sizes. It comes as a continuous **drop size distribution**, written `N(D)` — how many drops per cubic metre, per millimetre of diameter, at each size.

Our sums become integrals, and that is the only thing that changes:

$$Z = \int D^6 \, N(D) \, dD \qquad R = 6\pi \times 10^{-4} \int D^3 \, v(D) \, N(D) \, dD$$

...same two weightings. Same ambiguity. Just smoother.

20 GPM models $N(D)$ as a **normalised gamma distribution**:

$$N(D) = N_w \, f(\mu) \, (D/D_m)^{\mu} \, e^{-(4+\mu)D/D_m}$$

Look at what is blue. The two unknowns from the last page are the two parameters of the curve. Everything else is assumption.

21 And **μ**, the shape factor? The algorithm *assumes* it rather than retrieving it — there is not enough information to do otherwise.



So you've got one measurement, two unknowns, and a third parameter you're just... guessing?

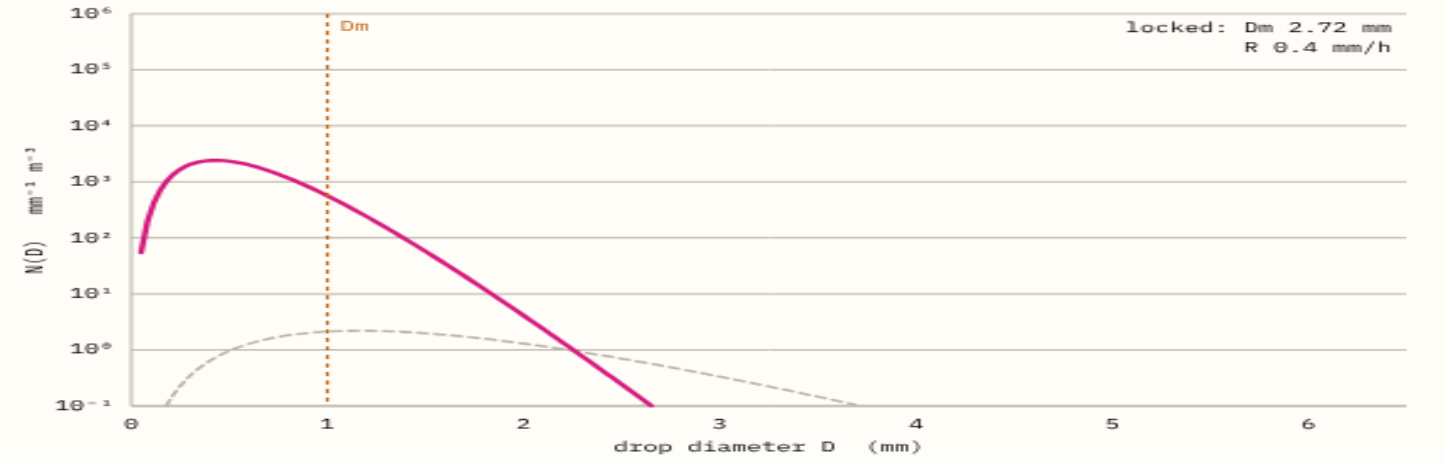
Yes. Which is why I carry a second radar. But first — play with the curve yourself.



Now You Try It: The Drop Size Laboratory

Drop size distribution laboratory

lock Z, then move D_m



D_m — mean drop size 1.00 mm

$\log_{10} N_w$ — how many 4.36

μ — shape (assumed, not retrieved) 3.0

Z locked at 29.0 dBZ | Box A | Box B | Reset

Z (RAYLEIGH)	RAIN RATE R	A IN Z=AR^1.4	LIQUID WATER
29.0 dBZ	3.79 mm/h	123	0.28 g/m³

What Core Actually Transmits: Details Part 7

(23) Two radars share the platform, and from here on they are colour-coded everywhere in this guide.

KuPR — 13.6 GHz, wavelength 22.0 mm. Essentially the TRMM design carried forward.

KaPR — 35.5 GHz, wavelength 8.4 mm. The new one, and the reason this whole guide has a happy ending.

(24)

PARAMETER	KUPR	KAPR
Frequency	13.6 GHz	35.5 GHz
Wavelength	22.0 mm	8.4 mm
Peak transmit power	1012 W	146.5 W
Beams per scan	49	49
Range resolution	250 m	250 / 500 m
Range sampling	125 m (oversampled)	125 / 250 m
Min. detectable rain	0.33 mm h^{-1}	0.19 mm h^{-1} (HS)
Z_{min} at L2 threshold	15.46 dBZ	13.71 dBZ (HS)

(25)



Wait. The Ka radar runs at a *seventh* of the power and still detects *lighter* rain than the Ku one? That can't be right.

It is right. At 35.5 GHz the same drops scatter far more efficiently — my wavelength is only four times your diameter. Hold that thought, because it is also about to cause me a great deal of trouble.



Chapter 4 is the trouble. Chapter 5 is why it was worth it.

Chapter 2: Self-Check

Radar reflectivity weights the drop size distribution by roughly D^6. Rainfall rate weights it by roughly which power?

- **A** $D^{3.7}$ — drop volume times fall speed, and fall speed itself goes as about $D^{0.7}$.
- **B** D^6 as well, which is why Z–R relations work so cleanly.
- **C** D^2 — rain rate is governed by the drops' cross-sectional area.
- **D** D^9 — rain rate is far more sensitive to large drops than reflectivity is.

Answer: A. Volume contributes D^3 and terminal fall speed about $D^{0.7}$, so R ∝ roughly the 3.7th moment. The gap between that and the 6th moment is exactly the room in which a single-frequency radar's rain rate becomes ambiguous.

In the laboratory panel, locking Z and then raising Dm made the rain rate fall sharply. Why?

- **A** Holding Z fixed while increasing drop size forces the concentration Nw down steeply, and R depends on concentration far more than Z does at a given size.
- **B** Larger drops fall more slowly, so they deliver less water per unit time.
- **C** The gamma distribution becomes unphysical at large Dm and the model saturates.
- **D** Attenuation grows with drop size, suppressing the retrieved rain rate.

Answer: A. $Z \propto Nw \cdot Dm^7$ and $R \propto Nw \cdot Dm^{4.7}$ (for this μ). Holding Z constant means Nw must fall as Dm^7; R then falls as $Dm^{4.7-7} = Dm^{-2.3}$. Same echo, very different rain. No attenuation involved — this is the ambiguity in the clean, unattenuated signal.

CHAPTER 03

Slicing the Storm!!!

Range is height. That one fact is what makes a radar worth putting in orbit — and it comes with a blind spot right where you least want one.

The Scan Curtain: Main Ideas

1. **The idea:** Core sweeps its antenna across the track, **49 footprints** per sweep, each about 5.5 km wide at nadir. The spacecraft's own motion advances the scan forward.

Stack enough sweeps and you have a three-dimensional volume of the storm.

2. Within each beam, the echo is chopped into **range bins** — 125 m apart, sampled from 19 km above the ellipsoid all the way down to the ground.

Most of those bins are empty air. The interesting ones are the handful that are not.

3. There is one problem, **Dot**, and it is at the bottom. The ground is by far the brightest thing in the scene — orders of magnitude brighter than you.





So the bottom of your curtain is ruined by the echo from the dirt?

Several hundred metres of it. More near the edge of the swath. And that is the layer where the rain actually reaches the ground.



*That layer has a name — the **clutter blind layer** — and Chapter 8 will show you why it is a serious problem over the Western Ghats.*

The Real Scan Pattern: Details Part 1

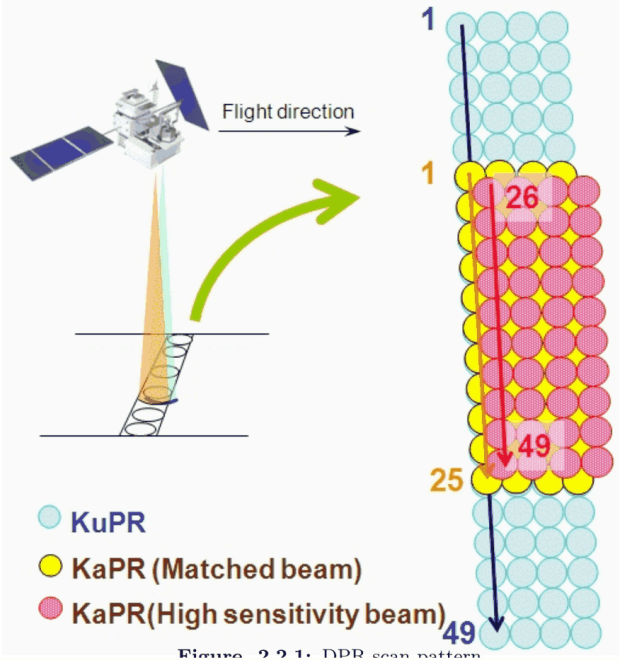


Figure 2.2.1, GPM/DPR Level-2 ATBD (Iguchi et al., rev. July 2024)

4 That is the real thing, from the ATBD. Three different sets of footprints, and they are not all the same radar.

- **Pale blue** — KuPR, 49 beams, the full swath.
- **Yellow** — KaPR **matched beams**, pointed at the same spots as the central 25 Ku beams so the two bands see the same volume of air.
- **Pink** — KaPR **high-sensitivity beams**, interlaced *between* the matched ones, with a longer pulse.

5



Why two Ka scan patterns? Pick one.

Because matching Ku exactly is what makes DFR possible — but a longer pulse on its own beams detects lighter snow. I could not have both in one beam, so I alternated.



6 On **21 May 2018** the KaPR scan pattern was changed so that Ka covers the **full swath**, not just the inner 25 beams. Dual-frequency retrieval became available edge to edge.

Remember that date. It is the single most important thing to know before you open a file, and page 7 of this chapter explains why.

Three Heights That Matter: Details Part 2

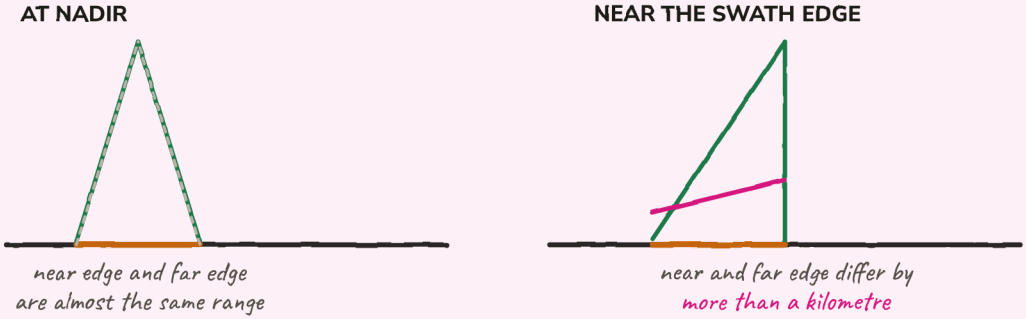
7 `binStormTop` — the highest bin with a detected precipitation echo. Above it, nothing.

It matters more than it looks: the forward retrieval *starts* here, so an error in storm top propagates down the whole profile as an attenuation error.

8 `binRealSurface` — the ground, found from the echo power profile together with the digital elevation model carried in the Level-1B file.

9 `binClutterFreeBottom` — the lowest bin Core is willing to trust. And it sits well *above* the surface.

The surface return leaks into nearby bins through the antenna's main lobe and, worse, through its side lobes. The effect gets dramatically worse away from nadir, and here is why:



A 5 km footprint seen at 17° off nadir has its near and far edges well over a kilometre apart in slant range, so the bright surface echo smears across many bins.

THE CONSEQUENCE!!!

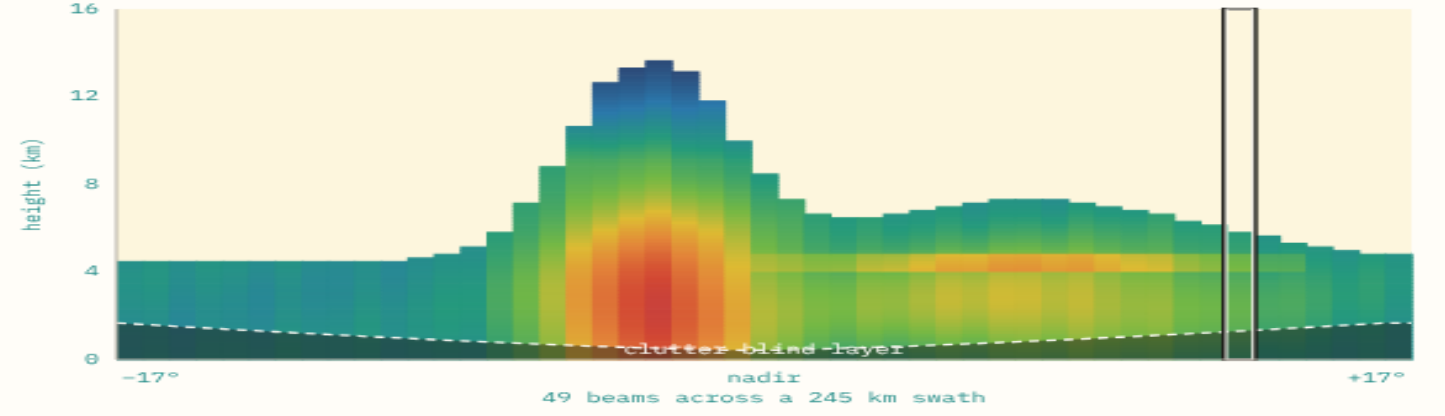
Everything below the clutter-free bottom is **extrapolated, not measured**. The Solver simply holds reflectivity constant from the lowest trusted bin down to the ground.

For deep convection that is defensible. For shallow rain over mountains it is a guess about the exact part of the profile you cared about.

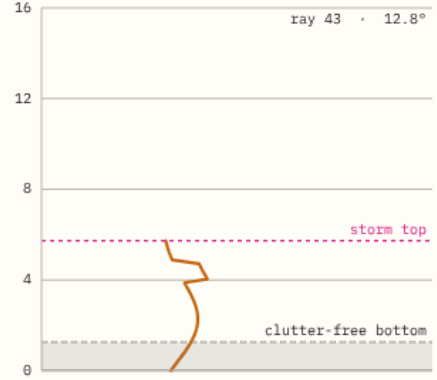
Now You Try It: The Scan Curtain

Scan curtain

click or drag across the curtain to pull out one ray







What Is Actually in the File: Details Part 3

(12) The curtain you just clicked on is, in the product, a set of arrays indexed by scan, angle bin, range bin and frequency. Learn these names early — they are in every GPM paper:

VARIABLE	WHAT IT IS
`zFactorMeasured`	the raw echo, before any correction
`zFactorNPCorrected`	after removing gas and cloud attenuation
`zFactorFinal`	fully attenuation-corrected — Chapter 4
`precipRate`	the answer
`paramDSD`	the retrieved N_w and D_m
`flagEcho`	whether that echo counted as precipitation

A TRAP!!!

Every echo above the noise floor is converted to reflectivity and written out — *including* echoes below the precipitation detection threshold.

`zFactorMeasured` having a value does **not** mean the algorithm considered it rain. `flagEcho` is what tells you that. Read the flags before you read the science fields.

The Shape of the File: Details Part 4

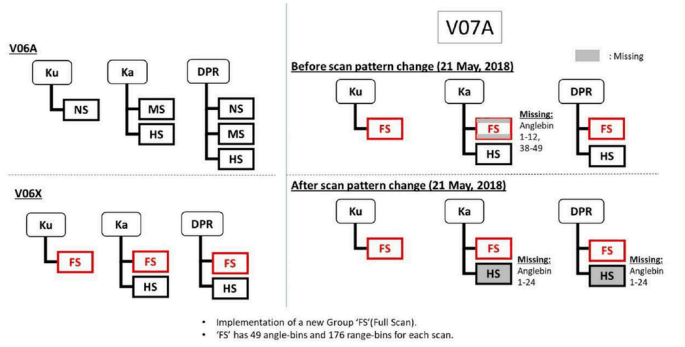


Figure 0.3.1, GPM/DPR Level-2 ATBD (Iguchi et al., rev. July 2024)

(14) An HDF5 file is a tree. The top branches of a DPR Level-2 file are **swaths**, and they were reorganised in Version 7.

- **V06** had three: `NS` (normal, Ku full swath), `MS` (matched, inner 25), `HS` (high sensitivity).
- **V07** has two: `FS` (full swath, dual frequency) and `HS`.

```
FS: 49 angle bins × 176 range bins, 125 m apart
```

THE TRAP IN THIS FIGURE!!!

V07 applies the `FS` format to data from *both sides* of the 2018 scan change — but the data behind it is not the same.

Before 21 May 2018: `Ka/FS` is missing angle bins **1–12 and 38–49**. **After:** `Ka/HS` is missing angle bins **1–24**. Same variable names, same shapes, silently different coverage.

Where Every Variable Lives: Details Part 5

16 Inside a swath, the variables are grouped **by the module that produced them** — the same seven modules of Chapter 7. Once you see that, the file stops being arbitrary.

GROUP	PRODUCED BY	WHAT YOU WILL REACH FOR
`ScanTime/`	—	`Year` , `Month` , `DayOfMonth` , `SecondOfDay`
`Latitude` , `Longitude`	—	per ray, at the swath level — not in a group
`PRE/`	Preparation	`zFactorMeasured` , `binStormTop` , `binClutterFreeBottom` , `binRealSurface` , `localZenithAngle` , `landSurfaceType` , `flagPrecip` , `elevation`
`VER/`	Vertical Profile	`binZeroDeg` , `heightZeroDeg` , `piaNP` , `attenuationNP`
`CSF/`	Classification	`typePrecip` , `flagBB` , `heightBB` , `widthBB` , `binBBPeak` , `flagShallowRain`
`DSD/`	Drop Size Distribution	`phase` , `binNode`
`SRT/`	Surface Reference	`pathAtten` , `reliabFlag` , `PIAalt` , `PIAweight`
`SLV/`	Solver	`precipRate` , `precipRateNearSurface` , `zFactorFinal` , `paramDSD` , `epsilon` , `piaFinal`
`FLG/` , `Experimental/`	various	quality flags and fields still being evaluated

TERMINOLOGY ALERT!!!

The fully attenuation-corrected reflectivity was called `zFactorCorrected` through V06. In **V07 it is** **`zFactorFinal`** .

Any script, tutorial or paper written before about 2022 uses the old name. If your read fails with a key error, this is why — and it is the most common first thing to go wrong.

Chapter 3: Self-Check

Why does the clutter-free bottom sit higher above the ground near the swath edge than at nadir?

- **A** At oblique incidence the surface return is spread across more range bins, because different parts of the footprint are at measurably different ranges.
- **B** The atmosphere is optically thicker along a slant path, so the lowest bins are attenuated away.
- **C** The antenna gain is lower at the scan edge, so the noise floor rises.
- **D** The spacecraft is farther from the surface at the edge of the swath, degrading range resolution.

Answer: A. A 5 km footprint viewed at 17° off nadir has a near edge and a far edge separated by well over a kilometre in slant range, so the bright surface echo smears over many bins — plus side-lobe returns. At nadir the whole footprint is at nearly the same range and the surface return is sharp.

What does the Solver do with the range bins below the clutter-free bottom?

- **A** It holds Ze constant at the value from the lowest trusted bin and carries that down to the surface.
- **B** It discards the pixel entirely and writes a missing value for surface rain rate.
- **C** It extrapolates the reflectivity gradient from the bins above, continuing the trend downward.
- **D** It substitutes the GMI radiometer's estimate for the lowest layer.

Answer: A. Constant Ze below the clutter-free bottom. That is a reasonable assumption for deep convection where the profile is already fairly uniform near the surface, and a much weaker one for shallow systems where the lowest kilometre is where the structure lives.

CHAPTER 04

The Signal Eats Itself!!!

The rain Core is trying to measure is also absorbing the pulse Core is measuring it with. At one of the two frequencies, it absorbs almost all of it.

Attenuation: Main Ideas

1



Small confession, **Core**. Every drop that scatters a bit of your pulse back to you also takes a bite out of the part that keeps going down.

How big a bite?





At Ku-band? Barely noticeable. At Ka-band, in a decent rain column, I will eat the whole thing.

2 **The quantity: specific attenuation** k, in decibels per kilometre. Like reflectivity it is a sum over the drops — but weighted by the *extinction* cross-section rather than the backscatter one:

$$k = 4.343 \times 10^{-3} \times \Sigma\, \sigma_e(D)$$

σ_e *in* mm^2*, summed over all drops in a cubic metre*

3 **The consequence:** what Core records at range *r* is the true reflectivity minus everything in front of it — twice over, because the pulse makes a round trip.

$$10 \log_{10} Z_m(r) = 10 \log_{10} Z_e(r) - 2 \int_0^r k(s)\, ds$$

measured *true* *everything it passed through*

That integral all the way to the ground has its own name: the **path-integrated attenuation**, or **PIA**.

Attenuation in Our Two Boxes: Details Part 1

4 Back to Box A and Box B. Here are the real extinction cross-sections, computed from Mie theory for liquid water at 10 °C:

DROP	Σ_e AT KU (MM²)	Σ_e AT KA (MM²)	KA ÷ KU
1 mm	0.031	0.331	10.7 ×
2 mm	0.872	7.028	8.1 ×
3 mm	5.941	21.856	3.7 ×

Psst! Notice the ratio shrinks as drops get bigger. Ka-band saturates — it is already near the limit of what a 3 mm drop can absorb — while Ku keeps climbing. Remember that; it is the seed of Chapter 5.

5 Add them up for **Box A** (one drop each):

$\Sigma\sigma_e$ Ku = 6.84 → k = **0.030** dB/km
$\Sigma\sigma_e$ Ka = 29.21 → k = **0.127** dB/km

6 And for **Box B** (794 drops of 1 mm):

$\Sigma\sigma_e$ Ku = 24.5 → k = **0.106** dB/km
$\Sigma\sigma_e$ Ka = 262.8 → k = **1.141** dB/km

The Tie-Breaker: Details Part 2

7 Now stack each box into a **4 km deep column** and work out the two-way attenuation to the ground. PIA = 2 × k × depth:

	Z (what core sees)	PIA Ku	PIA Ka	Difference
Box A — few, large	29.0 dBZ	0.24 dB	1.02 dB	0.8 dB
Box B — many, small	29.0 dBZ	0.85 dB	9.13 dB	8.3 dB

8 Look at the last column. The two boxes are **identical** in reflectivity — and the Ka channel is attenuated by **eight decibels more** in one than the other.

SPLAT!!!

9 So the attenuation is not only a problem. It is *information*. The thing that is ruining my Ka channel is also the thing that tells the two boxes apart.





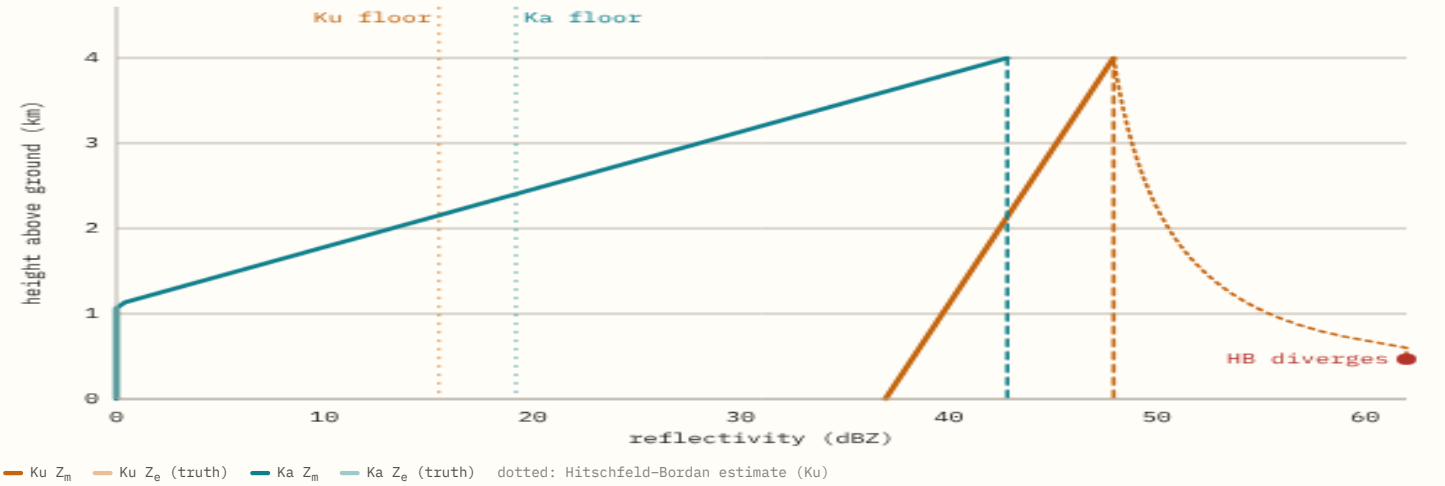
Now you're getting it. Chapter 5.

NOTE — be careful here. What separates these two boxes is mostly **water content**, not drop size: Box B holds twenty-two times as much liquid water, so it absorbs far more. Drop *size* leaves its own, separate fingerprint through non-Rayleigh scattering. The real algorithm uses both, and Chapter 5 keeps them apart.

Now You Try It: Attenuation Through a Column

Attenuation through a rain column

push the rain rate until Ka-band dies



Rain rate through the column — 28 mm h^{-1}

Rain layer depth — 4.0 km

D_m in the column — 2.30 mm

Show HB correction (Ku)

Hitschfeld–Bordan, and Why It Explodes: Details Part 3

12 The classical fix. Assume a power law between attenuation and reflectivity...

$$k = \alpha Z_e^{\beta}$$

...substitute it into the integral equation, and it solves in closed form:

$$Z_e(r) = Z_m(r) \div \left[1 - 0.2 \ln(10) \beta \int_0^r \alpha Z_m^{\beta} \, ds \right]^{1/\beta}$$

13 Now look only at the **denominator**. As the integrated attenuation grows, the bracket shrinks toward zero — and the correction goes to infinity.

So the formula works beautifully in light rain, where I did not need it, and detonates in heavy rain, where I did.



14 And there is a second, sneakier problem: α has to be **assumed before the retrieval starts**, because it encodes the very drop size distribution you are trying to find.

GO BACK AND TRY IT!!!

On the previous page, switch on the HB correction and drag $\mathbf{D_m}$ up past 2 mm. The simulator fits α and β once, to a Marshall–Palmer-like distribution — exactly what a climatological relation does — and then lets you move the real drops away from that assumption. Watch the mismatch readout climb past 1 and the denominator collapse.

Borrowing the Ground: Details Part 4

15 The way out is to measure the total attenuation **independently**, and Core has an elegant trick for it: use the ground.

The ground's radar cross-section, `σ⁰`, is a reasonably stable property of a given surface at a given incidence angle. So:

```
PIA = σ⁰( no rain ) − σ⁰( rain )
```

whatever dimmed the ground, the rain took out — twice.

This is the **Surface Reference Technique**.

16



Hang on. Where do you get the "no rain" measurement? It's raining. That's the whole situation.

From somewhere else. That is the entire art of the thing — and the reason it works far better over ocean than over your mountains.



17 The SRT module builds up to five estimates for each channel and weights them by the inverse of their variance:

REFERENCE	WHERE IT COMES FROM	WORKS OVER
Along-track	mean of the last 8 rain-free footprints at the same angle bin and surface type, run both forwards and backwards	everywhere
Cross-track	a quadratic fit of those averages across the 49 angle bins	`ocean only`
Temporal	climatological look-up tables of rain-free σ^{o} on a 0.5° grid, by angle and season, split across 5 surface types	everywhere

Why the cross-track fit is ocean-only: the sea surface is smooth and statistically uniform across a 245 km swath, so a quadratic through the angle bins is a good model of σ^{o}. Land is forest, then terrace, then bare rock, then reservoir, within a few kilometres. The fit does not behave, so it is simply not used.

*Every PIA estimate in the product comes with a **reliability flag**. Read it. The Solver certainly does.*

Chapter 4: Self-Check

What makes the Hitschfeld–Bordan solution unstable in heavy rain?

- **A** Its denominator contains the integrated attenuation and approaches zero, so any error in α or in calibration is amplified without bound.
- **B** It assumes Rayleigh scattering, which fails once drops become comparable to the wavelength.
- **C** It requires an independent PIA estimate, which the surface reference cannot supply over land.
- **D** It integrates upward from the surface, so ground clutter contaminates every gate above.

Answer: A. The bracket [1 − 0.2 ln10 · β ∫αZm^β ds] shrinks as attenuation accumulates. As it approaches zero the correction blows up, and a small bias in α — which is really a bias in the assumed DSD — becomes a catastrophic one. HB works where you do not need it and fails where you do.

How does the Surface Reference Technique estimate path-integrated attenuation?

- **A** By differencing the surface cross-section σ^0 measured inside the rain against a rain-free reference at the same incidence angle and surface type.
- **B** By comparing the Ku-band and Ka-band reflectivities at the lowest clutter-free bin.
- **C** By integrating the retrieved specific attenuation from storm top to the surface.
- **D** By using the GMI brightness temperature as an independent column constraint.

Answer: A. The ground is a target of roughly known brightness. Whatever dimming you see relative to its rain-free value is what the rain took out, two ways. The whole difficulty is establishing the rain-free reference — hence the along-track, cross-track and climatological estimates combined by inverse variance.

CHAPTER 05

Two Colours of Light!!!

One measurement, two unknowns. You cannot assume your way out of that. You need a second measurement — at a wavelength where the drops stop behaving.

Dual Frequency: Main Ideas

1

If my second radar behaved *exactly* like my first, the second measurement would be redundant and we would have gained nothing.





It doesn't behave the same, though. At 8.4 millimetres I'm not a small object any more.

2 **The idea in one line:** because the two bands disagree, and because *how much* they disagree depends on drop size, the **difference between them** measures size directly.

$$\mathrm{DFR} = 10\log_{10} Z(\mathrm{Ku}) - 10\log_{10} Z(\mathrm{Ka})$$

Dual-Frequency Ratio. A difference of logs, so really a ratio.

3 **And here is the elegant part**, which we will prove properly in two pages: in the ratio, the *concentration cancels out*.

DFR tells you D_m (and nothing about N_w)
then either reflectivity tells you N_w

Two measurements, two unknowns, system closed.

SPLAT!!!

...and then, on page 5, we will look at what the curve actually does, and the happy ending gets considerably more complicated.

When a Drop Stops Being Small: Details Part 1

4 Rayleigh scattering — the D^6 law from Chapter 2 — holds when a particle is **much smaller than the wavelength**. The usual yardstick is the size parameter:

```
x = π D / λ     — small x means Rayleigh
```

So for a 2 mm raindrop like Dot:

```
Ku:  x = π × 2 / 22.0 = 0.29   comfortably small
Ka:  x = π × 2 /  8.4 = 0.74   not small at all
```

5 Once x approaches 1, the scattering goes into the **Mie** regime. The cross-section stops following D^6, oscillates, and grows far more slowly.



In other words, at Ka-band I'm not a tiny dot any more — I'm a *lens*. And lenses are complicated.

NOTE: Every DFR number in this chapter is computed live from Mie theory for liquid water spheres at 10 °C — not from a fitted curve. The simulator validates against the Rayleigh limit to within 0.01 dB at small drop sizes.

6 That departure is not noise. It is **information** — and it is the only thing in this whole problem that responds to drop size and nothing else.

My Ku radar sees you the way physics-textbook Rayleigh says it should. My Ka radar sees you as you really are. The *disagreement* between them is my measurement.



Why the Concentration Cancels: Details Part 2

7 Four steps. Follow them in order and the result is unavoidable.

Step 1. Split the distribution into "how many" × "what shape":

$$N(D) = N_w \times n(D\ ;\ D_m)$$

↑ a number ↑ a curve whose shape depends only on Dm

Step 2. Reflectivity is an integral over N(D), and N_w is just a constant multiplier, so it comes straight out of the integral:

$$Z(Ku) = N_w \times I_{Ku}(D_m)$$
$$Z(Ka) = N_w \times I_{Ka}(D_m)$$

↑ these two differ — that is the whole point

Step 3. A difference of logs is the log of a ratio:

$$DFR = 10\ \log_{10} Z(Ku) - 10\ \log_{10} Z(Ka) = 10\ \log_{10} [\ Z(Ku) / Z(Ka)\ ]$$

Step 4. Substitute Step 2 into Step 3:

$$DFR = 10\ \log_{10} \frac{N_w \times I_{Ku}(D_m)}{N_w \times I_{Ka}(D_m)} = 10\ \log_{10} \frac{I_{Ku}(D_m)}{I_{Ka}(D_m)}$$

The N_w on the top and the N_w on the bottom are the same number. They cancel. Exactly, not approximately.

DOUBLE SPLAT!!!

8



So if you double the number of us without changing our sizes, both your echoes double, and the ratio doesn't move.

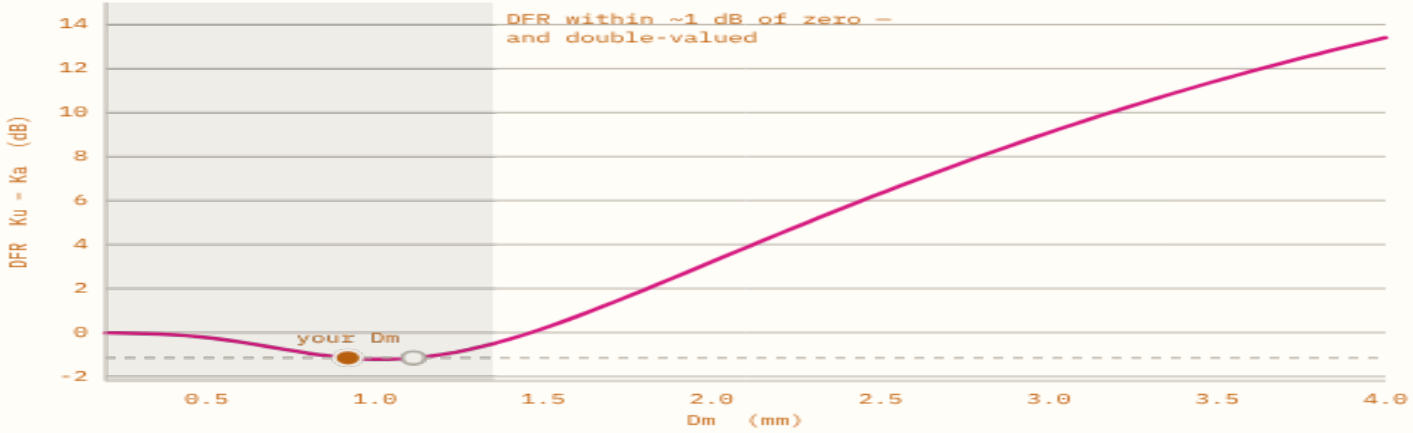
Exactly. Which means DFR is a pure size gauge. Go to the next page and try to break it — move the N_w slider as far as it goes and watch the DFR readout refuse to budge.



Now You Try It: DFR as a Size Gauge

DFR as a size gauge

Mie scattering, computed live



D_m 0.92 mm

$\log_{10} N_w$ — move it, watch DFR not care 4.00

μ 3.0

DFR (EFFECTIVE)	Z_E KU	Z_E KA	DFR SOLUTIONS
-1.15 dB	22.6 dBZ	23.8 dBZ	2

The Trough: Details Part 3

10 Here is the curve you were just moving along, written out:

D_M	0.2	0.5	1.0	1.5	2.0	2.5	3.0
DFR (dB)	−0.01	−0.23	−1.21	+0.19	+3.21	+6.32	+9.09

Above about 1.5 mm it climbs steeply and behaves exactly as advertised. Below that, it does something awkward: it sags **negative**, bottoming out near −1.2 dB at $D_m \approx 1.0$ mm, before coming back up.

11



Negative? Your Ka echo is *bigger* than your Ku echo? How?

Mie scattering is not a smooth decline. Around that size there is a brief resonance where Ka backscatter is *enhanced* relative to Rayleigh. It only lasts a decibel or so, but it lands right where most of the world's rain is.



12 **Two consequences, both bad:**

- Every DFR value between about **−1.2 dB and 0** corresponds to **two** different drop sizes — one below 1 mm, one above.
- The curve is so **flat** there that a few tenths of a decibel of calibration error moves your answer by a large fraction of a millimetre.

Light rain — most of the rain on Earth, and most of what falls over India during a break spell — lives in exactly that region.

TERMINOLOGY ALERT!!!

DFR and **DFR_m** are not the same thing, and papers are not always careful about which they mean.

DFR_m is the *measured* ratio — it contains both the scattering difference *and* all the accumulated differential attenuation from Chapter 4. **DFR** without the subscript usually means the *effective* or intrinsic ratio, after attenuation correction, which is the pure size gauge. The curve on this page is the effective one.

Which Way to Walk: Details Part 4

13 At each range bin we now have two equations and two unknowns — plus an attenuation term that has to come from somewhere. There are two places to start.

14 **FORWARD** — start at storm top, work down.

The attenuation above any bin is known, because you already solved the bins above it.

The cost: it inherits the Hitschfeld–Bordan instability from Chapter 4, **and** the double-valued trough from the last page — at every single bin.

15 **BACKWARD** — start at the surface, work up.

Now the *known* quantity is the attenuation below each bin, anchored by the SRT's independent PIA estimate.

The cost: you depend on the SRT being right, and on an assumption about the profile inside the clutter blind layer that nobody can check.

16 The ATBD is unambiguous about which one wins: the backward method is preferred because it generally gives **numerically stable and unique** solutions, where the forward method may have two solutions at each range bin.

In practice I do not commit to either. Chapter 7 shows what I actually do — I run the retrieval many times over and pick the version that best reconciles everything I know.



What Ka Buys Above the Freezing Level: Details Part 5

17 Rain is not the only reason Core carries a second frequency. Look at what DFR_m does in **snow**.

Ku reflectivity stays below about 25 dBZ — weak, as you would expect from low-density ice. And yet DFR_m runs several decibels, and **grows steadily downward**.

18



That's aggregation. On the way down, snowflakes collide and stick. They get *bigger*, so the two bands disagree more and more.

Which means I can estimate the **phase-transition height** — where ice becomes water — even in convective storms with no clear melting signature.



WHY THAT MATTERS BEYOND THE RADAR!!!

Every passive radiometer retrieval in the GPM constellation needs to know where the ice ends and the water begins — the two have completely different microwave signatures. Core's Ka channel is where that number comes from. The radar is not just measuring rain; it is handing the rest of the fleet a boundary condition.

TRIPLE SPLAT!!!

Chapter 5: Self-Check

Why is the effective dual-frequency ratio sensitive to Dm but not to Nw?

- **A** Both effective reflectivities scale linearly with Nw, so the concentration cancels exactly when you take their ratio.
- **B** Nw has no physical effect on backscatter; only drop size scatters radiation.
- **C** The Ka-band channel is calibrated against Ku-band, which removes the concentration dependence.
- **D** Nw cancels only in rain; in snow the DFR does depend on concentration.

Answer: A. $N(D) = Nw \cdot n(D; Dm)$, so Ze at either frequency is Nw times a function of Dm alone. Taking $10\log_{10}$ of the ratio subtracts the Nw terms away. That separation — size from the ratio, number from the absolute level — is the entire dual-frequency idea.

The ATBD prefers backward retrieval over forward retrieval. Why?

- **A** It generally gives numerically stable and unique solutions, where the forward method can have two solutions at each range bin.
- **B** It avoids needing a PIA estimate, which is unreliable over land.
- **C** It is faster, because the attenuation integral does not have to be recomputed at each gate.
- **D** It does not require knowing the storm top, which is poorly determined for shallow systems.

Answer: A. Backward retrieval starts from the surface with an independent PIA from the SRT, which anchors the solution. Its costs are real — it depends on the SRT being right and on an assumption about the profile inside the clutter region — but the forward method inherits both the HB instability and the DFR double-root at every gate.

CHAPTER 06

What Kind of Rain Is This?!!!

Before Core can choose a drop size distribution, it has to decide what it is looking at. The evidence is the shape of the profile itself.

Classification: Main Ideas

1. **Why bother?** Because the microphysics genuinely differs, and the algorithm carries **separate R–D_m relations** for the two regimes:

$$\text{stratiform:}\quad R = 0.401\,\varepsilon^{4.649}\,D_m^{6.131}$$

$$\text{convective:}\quad R = 1.370\,\varepsilon^{4.258}\,D_m^{5.420}$$

Classify a profile wrong and you have applied the wrong physics to the entire column.

2. **Three categories**, and the third one is the honest one:
   - **Stratiform** — widespread, gentle, layered.
   - **Convective** — vigorous, deep, lumpy.
   - **Other** — for echoes that are only cloud, or only noise, or that live entirely above the freezing level where the question has no good answer.

3. And the single most useful clue I have is, of all things, an **artefact**. A bright horizontal stripe in the echo that does not correspond to more rain at all.





Oh — the bright band. That's my sisters having a very bad few hundred metres.

The Bright Band: Details Part 1

④ Follow a single snowflake down through the 0 °C level. Six things happen in order, and only the last two bring the echo back down.

	WHAT HAPPENS	EFFECT ON THE ECHO
①	A large, loose snow aggregate falls slowly through the freezing level.	weak — ice scatters poorly
②	It starts to melt and acquires a **coat of liquid water**.	`↑↑ big jump`
③	Water's dielectric factor $\|K\|^2 \approx 0.93$; ice's is ≈ 0.18 — a factor of five.	`↑↑`
④	But it is still **snowflake-shaped and large** — it has not collapsed yet.	`↑↑ peak`
⑤	Melting completes. It collapses into a compact raindrop — much smaller.	`↓`
⑥	Now dense and compact, it **speeds up**, so fewer drops occupy each cubic metre.	`↓`

Briefly, at step ④, the particle is the worst of both worlds for a radar: the dielectric of water with the geometry of snow. Hence a **band**, not a step.

SPLAT!!!

⑤ Where to look for it: the CSF module searches a window from **4 bins above** the 0 °C range bin to **8 bins below** it.

*Experience from TRMM: the peak sits about **half a kilometre below** the freezing level.*

⑥ Why a bright band is near-conclusive evidence of stratiform rain:



A convective updraught is far too violent and turbulent to let a neat horizontal melting layer form. If you can see the band, nothing is stirring.

What a Melting Particle Actually Is: Details Part 2

7 Chapter 2 treated every particle as a sphere of water. The algorithm does not. It models each one as a sphere of **three things at once**:

$$P_w + P_i + P_a = 1$$

liquid water + solid ice + air, by volume

and tracks its bulk density ρ_b in g/cm³ as the mixture changes.

8

PHASE	P_w	ρ_b
Liquid	1	1.00
Solid	0	≈ 0.10
Mixed	in between	see below

*A snowflake at ρ_b = 0.10 is **ninety percent air**. Hold on to that — it is why snow is so hard.*

9 For a **mixed-phase** particle the ATBD sets the water fraction from a melting model, then gets the density from a startlingly simple empirical rule:

$$\rho_b = \sqrt{P_w}$$

That square root is doing a lot of work. Run it:

P_w — fraction melted by volume	0.01	0.25	0.64	1.00
ρ_b — density (g/cm³)	0.10	0.50	0.80	1.00



A quarter melted and it is already *half* the density of water?

The meltwater runs into the skeleton and collapses it. The structure goes long before the ice does. That is why a bright band is a few hundred metres thick and not a few kilometres.



SPLAT!!!

Mixing the Dielectric: Details Part 3

10 The dielectric constant of the mixture comes from a **mixing rule** — not an average, but one weighted the way scattering actually responds:

$$(\varepsilon_b - 1)/(\varepsilon_b + U) = P_w(\varepsilon_w - 1)/(\varepsilon_w + U) + P_i(\varepsilon_i - 1)/(\varepsilon_i + U)$$

$$U = 12.0 \qquad \text{if } \rho_b \le 0.09$$

$$U = 2.0 \cdot e^{13(\rho_b - 0.09)} \qquad \text{if } \rho_b > 0.09$$

A quirk worth knowing. Evaluate U on both sides of the boundary: at $\rho_b = 0.09$ it is **12.0**, and a hair above it is **2.0**. The published formula *steps* there — two different regimes for fluffy aggregates and for denser particles, joined without a smooth transition. It is in the document as printed; if your own forward model disagrees with GPM at very low densities, look here first.

Why the Echo Falls Back: Details Part 4

11 Now follow **one particle** whose melted mass would make a 1 mm raindrop, all the way down. Two things change together.

First its **size**. A low-density particle holding a fixed mass must be bigger:

$$D_b = D_{melt} \cdot \rho_b^{-1/3}$$

Then its **fall speed**, which the ATBD gives in three density regimes:

$$V_b = 3.3\ \sqrt{(\rho_b - \rho_a)} \qquad \rho_b \le 0.05$$

$$V_b = 8.8\ \sqrt{(0.1\ D_b(\rho_b - \rho_a))} \qquad 0.05 < \rho_b \le 0.30$$

$$V_b = \text{interpolate to the raindrop speed} \qquad 0.30 < \rho_b \le 1.00$$

12 Put the two together for our one particle:

ρ_b	D_b (mm)	V_b (m/s)	WHAT IT IS
0.05	2.71	0.73	a loose aggregate, drifting
0.10	2.15	1.28	a snowflake
0.30	1.49	1.86	wet, collapsing
0.50	1.26	2.58	half melted
1.00	1.00	3.78	a raindrop

It **shrinks to a third of its diameter** and **speeds up five-fold**, carrying exactly the same water.

Conserving the Flux: Details Part 5

(13) Here is the step that closes the argument — the **N/N model**, for *non-breakup, non-coalescence*. If no particle splits or merges on the way down, then the number crossing each level per second must be the **same at every level**. Speed up, and the concentration has to fall:

$N(D_b) \propto V_R(D_{melt}) / V_b(D_b)$

go three times faster, and there are a third as many of you per cubic metre



So by the time I am a raindrop, I am **a third the diameter** and there are **three times fewer** of me in each cubic metre than there were snowflakes.

Which is precisely why my echo collapses below the bright band. Not because the rain stopped — because the targets got small and sparse at the same time.



DOUBLE SPLAT!!!

Three Methods, One Verdict: Details Part 6

14 **THE V-METHOD** — vertical. Looks down a single beam.

```
bright band found   AND Z < 46 dBZ → stratiform
no bright band      AND Z > 40 dBZ → convective
neither                            → other
```

Two different thresholds, because a bright band is itself strong evidence — it earns a higher bar before the profile is called convective.

15 **THE H-METHOD** — horizontal. Ignores the vertical structure entirely and looks at the **pattern across neighbouring pixels**, using a modified Steiner–Houze–Yuter separation.

A pixel much brighter than its surrounding background is convective, whatever its own profile looks like.

This catches the embedded convective cores inside an otherwise stratiform deck, which the V-method would smooth over.

16 **THE DFR_m METHOD** — dual-frequency, and a genuinely different kind of evidence.

IN THE RAIN LAYER	DFR_M BEHAVIOUR	WHY
Stratiform	`small, flat with height`	Ku stays under ~25 dBZ, so there is almost no Ka attenuation to accumulate
Convective	`large, steep slope`	differential attenuation piling up fast, plus non-Rayleigh scattering off big drops

17



Why does the method use the *slope* of DFR_m rather than how big it is? Surely the actual value is more informative.

Because any fixed calibration offset between my two radars adds the *same constant* at every height. Differentiating with height removes it completely. Slope is free of a bias I cannot otherwise measure.



DOUBLE SPLAT!!!

Real Profiles, Real Data: Details Part 7

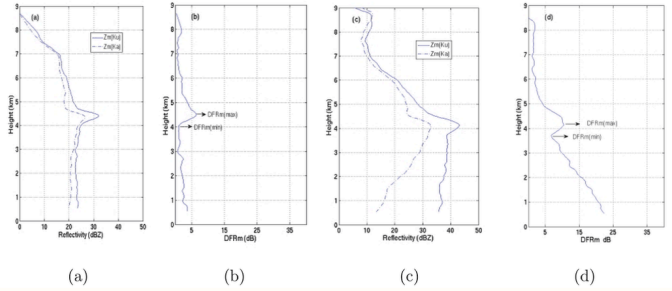


Figure 3.5.6, GPM/DPR Level-2 ATBD — airborne radar observations. (a),(b) stratiform; (c),(d) convective

18 Those are not simulations. Panels (a) and (b) are **stratiform**, (c) and (d) **convective**, measured from an airborne precipitation radar — and every claim on the previous page is visible in them.

- In (a) the bright band is the obvious bump near 4 km, and below it **DFR_m in (b) is flat and small** — barely 5 dB, barely changing with height.
- In (c) the convective profile has no such bump, and **DFR_m in (d) climbs steadily all the way down**.

The marked $DFR_m(max)$ and $DFR_m(min)$ are the quantities the classifier actually measures: their ratio becomes the index V1, and the mean slope below the minimum becomes V2.

GPM's First Snow: Details Part 8

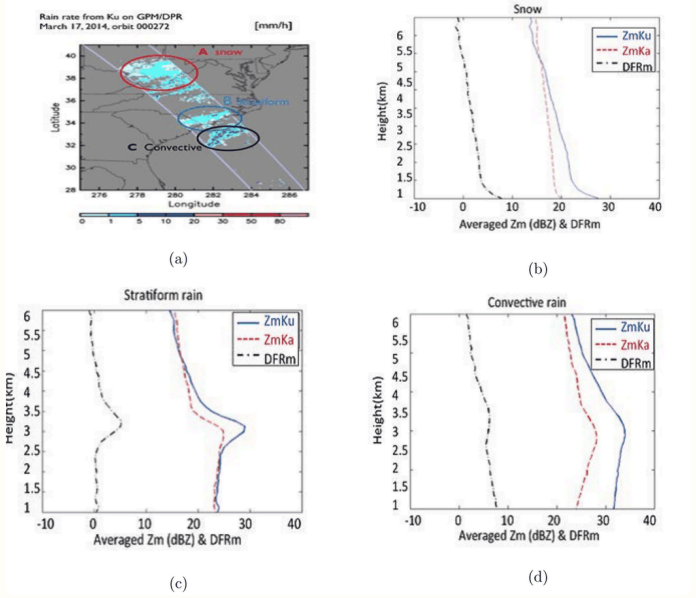


Figure 3.5.19, GPM/DPR Level-2 ATBD — GPM DPR orbit #000272, 17 March 2014

19 And this one is **GPM itself**, three weeks after launch — overpass #000272 over the eastern United States, catching snow, stratiform rain and convection in a single 800 km stretch of one orbit.

REGION	KU REFLECTIVITY	DFR_M BEHAVIOUR
A — snow	`mostly under 25 dBZ`	several dB, **growing downward** — aggregation, not attenuation
B — stratiform	`bright band at ~3 km`	near **zero and flat** below the melting layer
C — convective	`30–35 dBZ`	**large and sloping** throughout the rain

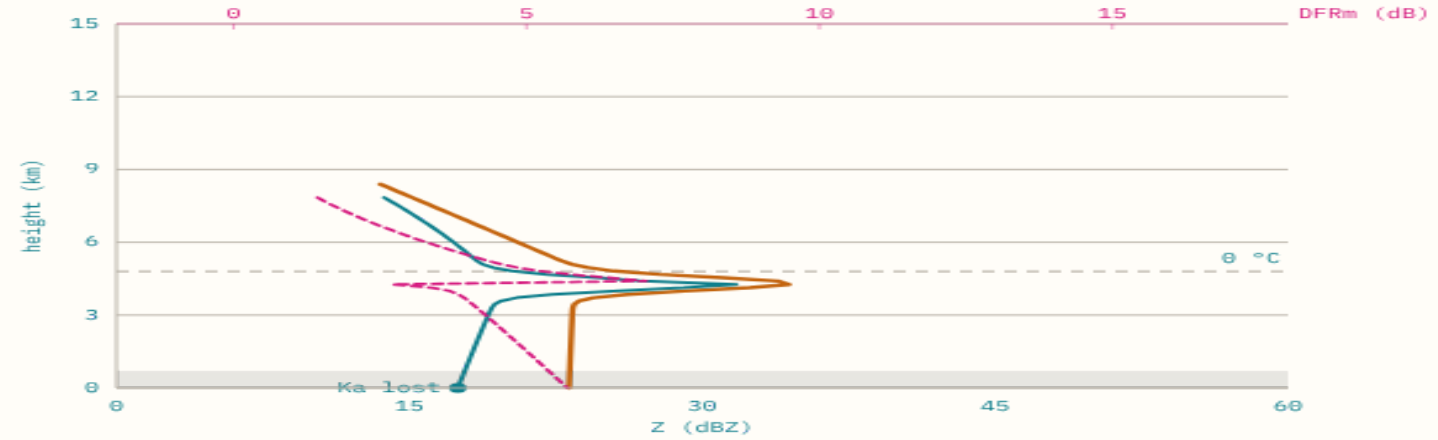
Three regimes, three completely different DFR_m signatures, one overpass. This single figure is why the dual-frequency classifier works.

TRIPLE SPLAT!!!

Now You Try It: Classify the Profile

Classify the profile

read the evidence, then commit



Z MAX (KU, MEASURED)	STORM TOP	BB PEAK	DFR_M SLOPE
34.5 dBZ	**8.4** km	**4.3** km	0.51 dB/km

Profile **1** of 6. What is it?

Stratiform | Convective | Snow | Shallow / warm

Next profile → 0 correct

Nodes A to E, and the Phase Integer: Details Part 9

21 All of this has to be written down per range bin, and the DSD module does it by placing **five nodes** through the profile:

NODE	WHERE
A	the bin nearest **−20 °C** — at and above it, Solid
B	bright band **top**
C	bright band **peak**
D	bright band **bottom** — at and below it, Liquid
E	the bin nearest **+20 °C**

Between B and D the particle temperature is pinned to **0 °C** whatever the air is doing. Between A and D the phase is Mixed.

22 When there is **no bright band** — convective, or stratiform without one — B, C and D all collapse onto the single 0 °C bin.

That is the whole difference in code: a melting layer with thickness, or a melting layer with none.

And it is why Chapter 7 insists the classifier runs *before* the microphysics. Those five bins are the output of CSF and the input to everything that follows.



23 The per-bin result is stored in `DSD/phase`, and its encoding is one of the neatest tricks in the product — a **single integer carrying both phase and temperature**:

VALUE	MEANS	EXAMPLE
`≥ 200`	**Liquid**, temperature = value − 200 °C	`210 → liquid at +10 °C`
`100–200`	**In the bright band**: 100 = top, 150 = peak, 200 = bottom	`125 and 175 are the only intermediates used`
`< 100`	**Frozen or part-frozen**, temperature = value − 100 °C	`80 → ice at −20 °C`

READ THIS BEFORE YOU PLOT IT!!!

`phase` is **not** a continuous field. Plotting it on a colour scale produces something that looks meaningful and is not — a jump from 99 to 200 is a change of 1 °C across the melting boundary, not a change of 101. Decode it into phase and temperature first, then plot those separately.

The Flags Beyond Three Types: Details Part 10

24 Version 7 carries far more structure than stratiform / convective / other, and the extra flags are usually where the interesting science is.

FLAG	WHAT IT MARKS
`flagShallowRain`	shallow isolated and non-isolated systems, flagged independently of rain type
`flagBB` , `heightBB` , `widthBB`	bright band presence, height, and width (geometrically corrected for beam width and zenith angle)
`flagHeavyIcePrecip`	strong ice signatures aloft
`flagGraupelHail` , `flagHail`	graupel and hail; the hail flag is new in V7 and dual-frequency only
`flagSurfaceSnowfall`	built from a snow index combining DFR_m slope, Ku maximum and storm top height
`typePrecip`	an **eight-digit number**, not a small integer

TERMINOLOGY ALERT!!!

`typePrecip` catches people out constantly. It is not 1, 2, 3. The **leading digits** give the major category; the **trailing digits** record which method decided it and on what evidence.

If you write `typePrecip == 2` you will match nothing at all. Divide down to the leading digit, or consult the user manual's bit definitions.

26



So you've classified me. Now what?

Now I finally solve for you. But classification was only one of six things that had to happen first — and the next chapter is the whole assembly line.



Chapter 6: Self-Check

What physically produces the bright band?

- **A** Melting snow aggregates acquire a water coating — and water's dielectric constant — while still holding their large, loose geometry, before they collapse and accelerate into raindrops.
- **B** Updraughts concentrate hydrometeors into a thin layer just below the freezing level.
- **C** The radar beam refracts at the 0 °C level, focusing energy into a narrow range of heights.
- **D** Supercooled water droplets freeze suddenly at 0 °C, releasing latent heat that enhances backscatter.

Answer: A. A wet, large aggregate is close to ideal as a radar target: water's $|K|^2$ is roughly five times ice's, and the particle has not yet shrunk. Once melting completes, the particle collapses to a much smaller raindrop and speeds up — fewer, smaller scatterers per volume — and the reflectivity falls back. Hence a band, not a step.

Why does the DFRm classification method key on the slope of DFRm rather than its absolute value?

- **A** Slope is insensitive to a calibration offset between the two radars, which would shift absolute DFRm but not its vertical gradient.
- **B** Absolute DFRm is unavailable in the outer swath, where only Ku-band observes.
- **C** The slope is proportional to rain rate, so it converts directly to a precipitation estimate.
- **D** Absolute DFRm saturates above 10 dB, so it carries no information in heavy rain.

Answer: A. Any fixed bias between KuPR and KaPR calibration adds a constant to DFRm at every height. Differentiating with height removes it. The ATBD states this explicitly as the reason for using the slope in the snow index as well.

CHAPTER 07

Seven Modules!!!

Everything so far assembles into one processing chain. Six modules exist only to set up the equation that the seventh one solves.

The Level-2 Chain: Main Ideas

1 A main module calls **seven sub-modules** in order. They cannot call each other. Each one hands its result back and stops.

	MODULE	WHAT IT ADDS TO THE RAY
1	**PRE**	echo power → Z_m; rain / no-rain; storm top, surface, clutter-free bottom
2	**VER**	removes gas and cloud attenuation; flags the 0 °C bin
3	**CSF**	finds the bright band; classifies the rain type
4	**DSD**	assigns phase per bin; selects the N(D) and R–D_m relations
5	**SRT**	path-integrated attenuation from the surface echo
6	**SLV**	**the retrieval** — everything above was preparation
7	**TRG**	trigger variables, new in V7

2 **Three algorithms share this framework:** Ku-only, Ka-only, and the dual-frequency DPR algorithm that takes both single-frequency products as its input.

The two single-frequency runs go first, in either order. The dual-frequency run goes last.

Each processes one orbit at a time — but a whole month has to be processed as a unit for the databases to come out right.

3 **The ordering is not arbitrary:**

- **CSF before DSD** — the drop size assumptions depend on rain type and bright-band position.
- **DSD before SLV** — the Solver needs the scattering tables DSD selected.
- **SRT anywhere** between VER and SLV — it only needs clean surface cross-sections.

The Real Diagram: Details Part 1

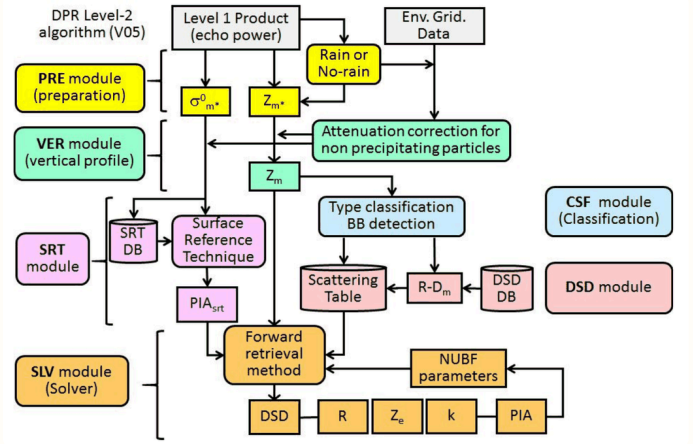


Figure 3.1.1, GPM/DPR Level-2 ATBD (Iguchi et al., rev. July 2024)

4 That is the actual flow chart, colour-coded by module. Follow it from the top and every box is something you have already met:

- **Level 1 (echo power)** enters at the top; **PRE** turns it into σ^0_m and Z_m and decides rain or no-rain.
- **VER** (green) takes the environmental grid and strips the non-precipitating attenuation, giving clean Z_m.
- **SRT** (pink) reads its own database to produce PIA_{srt}.
- **CSF** (blue) does type classification and BB detection — and feeds the **R–D_m** choice.

5 Two things in that diagram we have not met yet.

The **cylinders** are databases, not code: the SRT σ^o climatology and the DSD database holding the ε prior. They are updated offline, between processing runs, which is why a whole month must be processed as a unit.

And at the bottom right: **NUBF parameters**.

NON-UNIFORM BEAM FILLING!!!

Core's footprint is about **5 km across**. Rain is not uniform over 5 km — a convective core three kilometres wide sits inside one footprint alongside much lighter rain.

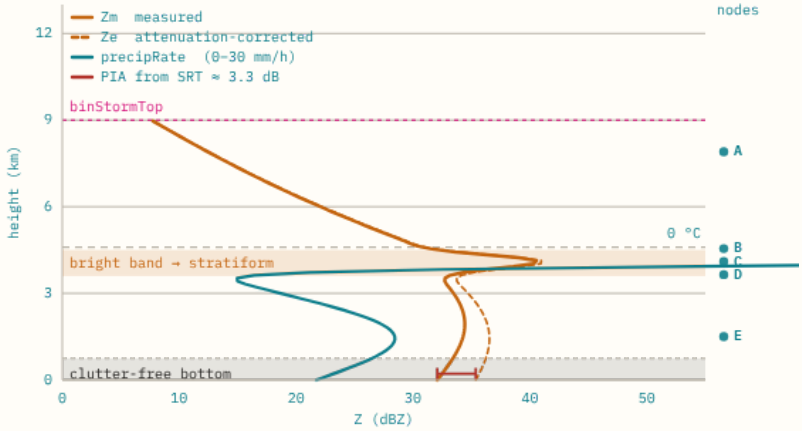
Because reflectivity and attenuation are **non-linear** in rain rate, averaging over a non-uniform beam does not give the average rain rate. It biases the answer, and the bias grows with how lumpy the rain is.

That is what the NUBF correction in the diagram is for, and it is one of the reasons heavy convective rain remains the hardest case even with two frequencies.

Now You Try It: One Ray Through the Chain

One ray through the Level-2 chain

step forward and watch the profile change

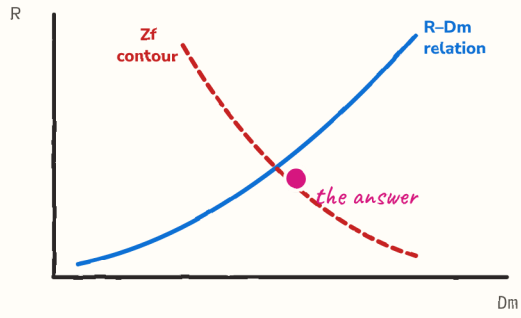


- Level-1B
- 1 PRE — Preparation
- 2 VER — Vertical Profile
- 3 CSF — Classification
- 4 DSD — Drop Size Distribution
- 5 SRT — Surface Reference
- **6 SLV — Solver**
- 7 Output

The R–D_m Plane: Details Part 1

8 The Solver works on a two-dimensional plane: **rain rate** up the vertical axis, **mass-weighted mean diameter** across the bottom. Every possible answer for a range bin is a point on it.

Two things are drawn on that plane, and the answer is where they cross.



One crossing is the happy case. The next step is what happens when there are two — or none.

9 The two curves:

- **The R–D_m relation** — an assumption, of the form $R = \varepsilon^{I} \, p \, D_m^{q}$, chosen by the DSD module from the rain type.
- **The Z_f contour** — every (R, D_m) pair that would produce the reflectivity actually observed at this bin, drawn from the Mie scattering tables.

CROSSINGS	WHAT THE SOLVER DOES
one	take it. Done.
two	prefer the solution with the **smaller** R and D_m
none	take the closest point on the R–D_m curve, and **record the discrepancy** — it becomes a cost term on the next page

The Epsilon Sweep: Details Part 2

10



That blue curve is an *assumption* though. What if the assumption is wrong for my particular storm?

It usually is. So I do not assume one curve. I try **a whole family of them** and keep the one that fits everything else I know.



11 The family is indexed by a single adjustment factor, **ε**. The Solver runs the entire retrieval repeatedly, sweeping

```
ε from 0.2 to 5.0
```

and picks the value that minimises a cost. In V7, ε is even allowed to **vary with range** rather than being fixed for the whole profile.

12 The cost has four terms. Three for the single-frequency algorithms, all four for dual-frequency:

	TERM	IN PLAIN WORDS
a	prior on ε	"don't wander too far from the climatology in the DSD database"
b	$PIA_{SLV} - PIA_{SRT}$	"the attenuation you produced had better match what the ground says" — one-sided when the SRT is saturated
c	$Z_{f1} - Z_{f2}$	"and the answer should actually reproduce the reflectivity you were given"
d	Ka consistency	"if you solved from Ku, the result must also explain the Ka observation"

SPLAT!!!

That is the whole retrieval. It is not a formula — it is a ***constrained optimisation*** *run independently for every pixel in every orbit.*

What This Means When You Use the Data: Details Part 3

(13) **Consequence one.** `precipRate` carries **different effective uncertainty in different regimes** — tight where the SRT is reliable and the DFR is informative; loose over land, in light rain, where neither is.

A single bulk error bar for a whole product is close to meaningless.

(14) **Consequence two.** The `epsilon` field is a **diagnostic**, not just an internal variable.

It tells you how far the algorithm had to move from its climatological DSD prior to reconcile the observations. A systematically non-unity ε over your region is telling you something physical about the microphysics there.

THE HABIT TO FORM!!!

When a retrieved value surprises you, do **not** treat `precipRate` as a measurement. Walk back up the chain:

`precipRate` → `zFactorFinal` → `pathAtten` and its reliability flag → `typePrecip` → `flagEcho`. Every stage of a seven-stage inference is visible in the file. Use that.

One chapter left, **Dot**. And it is the one where everything I have told you starts going wrong at the same time.





The monsoon. Of course it is.

Chapter 7: Self-Check

Why must the Classification module run before the DSD module?

- **A** The DSD module chooses particle densities, dielectric constants, fall speeds and the R–Dm relation according to rain type and bright-band position, which CSF determines.
- **B** CSF computes the path-integrated attenuation that DSD needs for its scattering tables.
- **C** DSD can only process pixels that CSF has already identified as containing precipitation.
- **D** The order is arbitrary; the modules are independent and can run in any sequence.

Answer: A. The two R–Dm relations differ substantially between stratiform and convective, and the node placement A–E that sets each bin's phase depends on where the bright band is. Get the type wrong and the whole microphysical parameterisation for that ray is wrong.

What is the parameter ε doing in the Solver?

- **A** It adjusts the R–Dm (equivalently k–Ze) relation, and the Solver sweeps it from 0.2 to 5.0 to find the value that best reconciles the DSD prior, the SRT PIA and the reflectivity residuals.
- **B** It is the fractional error in the radar calibration constant, estimated once per orbit.
- **C** It is the shape parameter μ of the gamma distribution, retrieved at each range bin.
- **D** It scales the non-precipitation attenuation correction supplied by the VER module.

Answer: A. ε is the knob that lets the algorithm depart from its climatological DSD assumption when the observations demand it. Because it is selected by a cost function rather than computed, the retrieved ε field is a useful diagnostic: a systematic departure from unity over a region is telling you the prior does not fit that region's microphysics.

CHAPTER 08

The Monsoon Problem!!!

Over the Western Ghats, every assumption in this guide gets tested at once — and Box A and Box B turn out to be real places, in real seasons.

Over India: Main Ideas

1. **The good news first.** GPM's 65° inclination covers the whole subcontinent, and the non-sun-synchronous orbit from Chapter 1 samples the monsoon's strong diurnal cycle properly — afternoon convection over the plains, nocturnal convection offshore, at different local times across a season.

2. **The bad news.** Three things go wrong, and they are the three things this guide has been building toward:

- the drop size distribution **moves** between monsoon phases;
- the rain that matters most falls **inside the blind layer**;
- the surface reference **degrades over land**.

3.



Here's the thing, **Core**. In an active spell I'm a big drop in a deep moist column. In a break spell I'm a small drop in dry air. *Same mountain, same season.*

...so a single fixed Z–R relation has to be wrong in at least one of those phases.





Both, probably. Let me show you the numbers.

Box A and Box B Were Real: Details Part 1

4 Disdrometer observations over the **Western Ghats** show the drop size distribution shifting systematically between the active and break phases of the Indian summer monsoon:

PHASE	mean D_m	mean $\log_{10} N_w$	modal $\log_{10} N_w$	MICROPHYSICS
Wet spell (active)	1.30 mm	3.62	3.9	collision–coalescence in a deep moist column: midsize and large drops
Dry spell (break)	0.92 mm	4.46	3.9 *and* 5.0 bimodal	evaporation and breakup in drier air: many more small drops

SOURCE. Joss–Waldvogel disdrometer at Mahabaleshwar (17.92° N, 73.6° E), JJAS 2012–2015; spells from IMD rainfall anomalies. Murali Krishna *et al.*, *Atmos. Chem. Phys.* **21**, 4741–4757 (2021). The two N_w columns are different quantities: the **mean** is what you compute with, the **mode** is where the histogram peaks — and the dry-spell histogram has two peaks. Note also that dry spells do not rain *harder*: mean R is 6.62 mm/h wet against 2.79 dry. What moves is the echo-to-rain relationship.

5 Now put those side by side with the two boxes you built by hand in Chapter 2:

	D_M	N_W	CHARACTER
Box A (invented)	2.72 mm	low	few and large
Wet spell (measured)	1.30 mm	lower	few and large
Box B (invented)	1.00 mm	high	many and small
Dry spell (measured)	0.92 mm	higher	many and small

Your toy boxes were not a thought experiment. They were the Western Ghats in July and the Western Ghats in August.

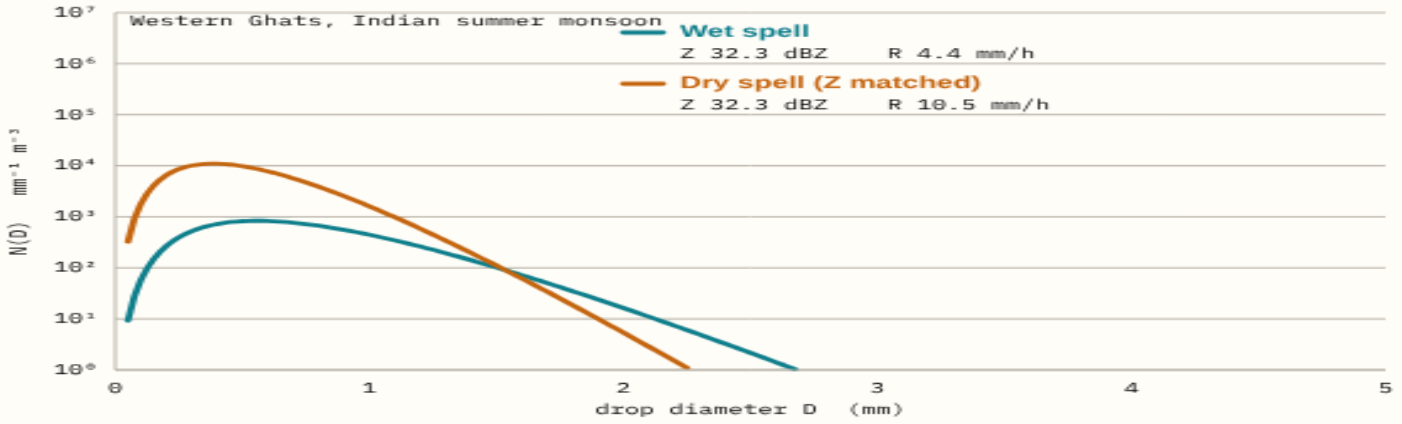


SPLAT!!!

Now You Try It: Wet Spell vs Dry Spell

Wet spell vs dry spell, same radar

measured Western Ghats parameters



Wet spell | Dry spell | Match Z

Readouts below: Dry spell (Z matched).

D_M	$LOG_{10}\ N_W$	Z	TRUE R	R FROM Z=200R^1.6	ERROR
0.90 mm	5.02	32.3 dBZ	10.5 mm/h	3.8 mm/h	-63 %

6 **Press "Match Z".** Both spells are set to the same reflectivity — the same thing a single-frequency radar would measure — and the true rain rates still differ by a factor of more than two.

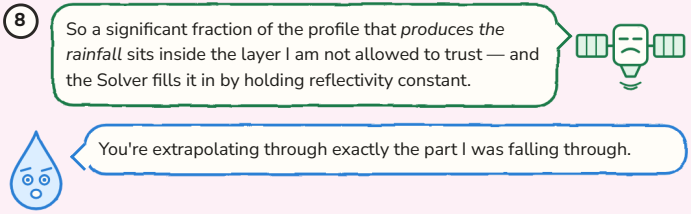
At any fixed reflectivity, dry-spell microphysics implies about **2.3 times** the rain rate of wet-spell microphysics — and that factor barely moves with rain rate, because at fixed D_M both Z and R scale with concentration. No single $Z = aR^b$ can sit on both.

Rain That Falls in the Blind Spot: Details Part 2

7 The windward slopes of the Ghats produce enormous accumulations from relatively **shallow warm-rain systems** forced up a steep escarpment. Those systems can be only two or three kilometres deep.

Now recall Chapter 3's clutter-free bottom: a few hundred metres above the surface at nadir, over a kilometre near the swath edge — and higher still over rough terrain, where the surface return is messier.

8



9 **And the surface reference degrades too.** The cross-track fit from Chapter 4 is ocean-only. Over the Ghats the along-track reference must be drawn from rain-free footprints at the same incidence angle over a surface that changes character within a few kilometres.

PIA reliability drops exactly where attenuation correction matters most — and cost term **(b)** from Chapter 7 weakens with it.

AND THE PRIOR FIGHTS YOU!!!

Cost term **(a)** anchors ε to a prior from the DSD database. If that prior reflects a global or tropical-mean microphysics, and the Western Ghats in a break spell is running a bimodal, small-drop distribution, then the algorithm is being pulled toward the wrong answer **by its own regularisation**.

The retrieved `epsilon` field over the region, averaged by monsoon phase, is a direct measurement of that pull. It is the first thing worth plotting.

How This Gets Checked: Details Part 3

(11) Validation over India leans on three independent sources:

- **IMD gridded gauge analysis** — dense over land, the usual reference for monsoon-season evaluation.
- **Ground radar and disdrometers** — these give the *actual* drop size distribution rather than an inference of it.
- **Inter-comparison** between DPR, IMERG and the earlier TMPA products over the monsoon core region.

(12) The consistent pattern in that literature: satellite estimates capture the **spatial structure and seasonal cycle** well, and struggle most with **heavy orographic events** and with **light rain**.

Two tails, two different reasons — and you now know both of them.

IF YOU WORK WITH THIS DATA!!!

Stratify before you compare anything. A bulk bias statistic over a whole season mixes wet-spell convection, break-spell shallow rain and stratiform decks — three regimes the algorithm handled with three different sets of assumptions.

The interesting signal is almost always in the stratification, not in the mean.

Opening a Granule: Details Part 4

14 A Level-2 granule is one orbit, in HDF5. Everything in this book is a few lines away:

```
import h5py, numpy as np

f  = h5py.File(path, "r")
fs = f["FS"]                              # the full-swath group

lat   = fs["Latitude"][:]                  # (nscan, 49)
lon   = fs["Longitude"][:]
rate  = fs["SLV/precipRate"][:]            # (nscan, 49, 176)
hgt   = fs["PRE/height"][:]                # (nscan, 49, 176), metres
ptype = fs["CSF/typePrecip"][:]            # (nscan, 49)
cfb   = fs["PRE/binClutterFreeBottom"][:]  # (nscan, 49), an index
```

15 Note `PRE/height`. The file **gives you the height of every range bin**, already corrected for the slant path and the local zenith angle.

Do not compute it yourself from bin numbers and 125 m. People do, and they get the swath edge wrong.

16 Before anything else, **print the shapes**. Some fields in the dual-frequency product carry a trailing frequency axis of length 2 — Ku first, then Ka — and some do not:

```
for k in ["PRE/zFactorMeasured",
          "SLV/zFactorFinal",
          "SLV/paramDSD"]:
    print(k, fs[k].shape)
```

It costs one second and saves an afternoon.

Six Things That Will Catch You: Details Part 5

17 **1. Range bin 0 is at the top.** Bin 0 is near 19 km; bin 175 is at the ground. Height *decreases* as the index increases, which is the opposite of every intuition.

18 **2. The fill value is −9999.9.** It is a large negative number, not a NaN. Average without masking it and your rain rates come out catastrophically negative.

```
rate = np.where(rate < -1000, np.nan, rate)
```

19 **3. `typePrecip` is eight digits.** Testing `== 2` matches nothing. The leading digit is the category:

```
major = ptype // 10000000   # 1 strat, 2 conv, 3 other
```

20 **4. `binClutterFreeBottom` is an index, not a height.** Use it to index the bin axis, and treat everything below it as extrapolated — Chapter 3.

21 **5. The May 2018 scan change.** Before it, `Ka/FS` has no data in angle bins 1–12 and 38–49. Any statistic you compute across the full swath will silently mix two different instruments unless you split on that date.

22 **6. `zFactorCorrected` became `zFactorFinal` in V07.** Every tutorial written before about 2022 uses the old name.

Five of these six are version or geometry traps, not science. That is the usual ratio.

A Curtain in Fifteen Lines: Details Part 6

(23) This is the Chapter 3 scan curtain, from a real granule — along-track this time rather than across:

```
import matplotlib.pyplot as plt

s0, s1, ray = 2000, 2120, 24          # 120 scans, nadir ray

z   = fs["SLV/zFactorFinal"][s0:s1, ray, :]       # (120, 176) dBZ
hkm = fs["PRE/height"][s0:s1, ray, :] / 1000.0   # (120, 176) km
cfb = fs["PRE/binClutterFreeBottom"][s0:s1, ray]

z = np.where(z < -1000, np.nan, z)

scan  = np.arange(s0, s1)
clut  = np.array([hkm[i, cfb[i]] for i in range(len(cfb))])

plt.pcolormesh(scan, hkm[0], z.T, vmin=15, vmax=50, shading="nearest")
plt.plot(scan, clut, "w--", lw=1.5)     # the blind-layer ceiling
plt.ylim(0, 15); plt.ylabel("height (km)"); plt.xlabel("scan")
```

(24) That dashed white line is the one to draw first, every time. Everything beneath it in my product is an assumption, not a measurement — and it is the part nearest the ground.





Plot the line. Then decide whether your answer depends on what is under it.

SPLAT!!!

Where to Go From Here

25 The Level-2 products live at NASA's PPS (`arthurhou.pps.eosdis.nasa.gov`) and JAXA's G-Portal, in HDF5. `2A.GPM.DPR` is the dual-frequency Level-2 file.

Everything this guide referenced is in there: `zFactorMeasured` , `zFactorFinal` , `precipRate` , `paramDSD` , `typePrecip` , `flagBB` , `binClutterFreeBottom` , `pathAtten` , `epsilon` .

26 **Two habits worth forming:**

- Read `flagEcho` and the quality flags **before** the science fields.
- When a number surprises you, walk back up the chain rather than arguing with the end of it.

NOTE: the GPM/DPR Level-2 ATBD is the authority for what each field means. The user manual is the authority for flag bit definitions — which this guide has deliberately not reproduced, because they change between versions.

27



So after all that — eight chapters, two radars, seven modules, a constrained optimisation — you still don't actually know how big I am.

No. But I know it to within a factor that is useful, over the whole planet, every three hours, including the seventy percent of it with no rain gauges. And I know **how** uncertain I am, which is the part most people forget to ask for.





...fine. That's quite good actually.

TRIPLE SPLAT!!!

Chapter 8: Self-Check

Why is a single fixed Z–R relation especially poor over the Western Ghats during the monsoon?

- **A** Dm shifts from about 1.3 mm in wet spells to about 0.9 mm in dry spells, so the same reflectivity corresponds to substantially different rain rates in the two phases.
- **B** Orographic rain is too heavy for the radar's dynamic range, saturating the receiver.
- **C** The 65° inclination means India is sampled only at the extreme edge of the swath.
- **D** Ground clutter from the mountains contaminates the entire profile, not just the lowest bins.

Answer: A. The published disdrometer record shows the distribution reorganising between active and break phases — collision–coalescence producing larger drops in wet spells, evaporation and breakup producing many small ones in dry spells. A relation fitted to one phase will be biased in the other, over the same mountains in the same season.

What is the main geometric problem for shallow orographic rain over the Ghats?

- **A** Much of the profile falls inside the clutter blind layer, so the layer actually producing the rainfall is extrapolated rather than observed.
- **B** The swath is too narrow to cover the escarpment in a single overpass.
- **C** Shallow rain falls below the DPR's minimum detectable reflectivity at both frequencies.
- **D** The bright band sits below the terrain, so classification always returns "other".

Answer: A. A warm orographic system can be only two or three kilometres deep, while the clutter-free bottom sits a few hundred metres up at nadir and over a kilometre up near the swath edge — higher still over rough terrain. The Solver then holds Ze constant through precisely the layer you wanted measured.